

Bachelor in Computer Applications

Course Title: **Digital Logics**

Code No: Semester: **1**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Define BCD code. Design BCD to Excess-3 Code converter with truth table and logic diagram.
2. Define combinational logic circuit. Design a combinational circuit whose input is a four-bit number and whose output is the 1's complement of the input number.
3. Define parallel counter. Design MOD-12 synchronous up counter along with state diagram, timing sequence and timing diagram.
4. Convert (591.62)₁₀ into hexadecimal and octal number system.
5. Why NAND and NOR gates are called Universal logic gates? Realize NOR as Universal logic gates.
6. Simplify: $F = X Y Z' + X Z' + X' Y + X'' Y Z'$ using K-Map in both SOP and POS.
7. Define PLDs. Implement the given functions with ROM. $F_1 = \sum(0,1,2)$ $F_2 = \sum(0,2,3)$
8. Describe the clocked master-slave J-K flip-flop with its operation table.
9. What is forbidden state in SR flip flop? Convert SR to JK flip flop.
10. Differentiate between ring counter and Johnson counter.
11. Explain the operation of 4-bit Parallel-In parallel-Out Shift register with data input 1010.
12. Write short notes on: (Any two) a. ASCII Code b. Logic micro operation c. State table